ENGINE 4.0L (Continued)

VALVES

DESCRIPTION	SPECIFICATION
Face Angle	46.5°
Head Diameter	
Intake	48.38 - 48.64 mm (1.905 - 1.915 in.)
Exhaust	37.97 - 38.22 mm (1.495 - 1.505 in.)
Length (Overall)	
Intake	122.47 - 122.86 mm (4.822 - 4.837)
Exhaust	122.86 - 123.24 mm (4.83 - 4.85 in.)
Stem Diameter	7.89 - 7.92 mm (0.311 - 0.312 in.)
Stem - to - Guide Clearance	0.025 - 0.076 mm (0.001 - 0.003 in.)
Tip Refinishing (Max. Allowable)	0.025 mm (0.010 in.)

VALVE SPRING

DESCRIPTION	SPECIFICATION
Free Length (Approx)	47.65 mm (1.876 in.)
Spring Force (Valve Closed)	
Intake and Exhaust	316.0 - 351.0 N @ 41.65 mm (71.0 - 79.0 lbs. @ 1.64 in.)
Spring Force (Valve Open)	
Intake and Exhaust	898.6 - 969.7 N @ 30.89 mm (202.0 - 218.0 lbs. @ 1.216 in.)
Wire Diameter (Inside) Intake and Exhaust	21.0 - 21.51 mm (0.827 - 0.847 in.)
Installed Height (Spring Seat to Bottom of Retainer)	41.65 mm (1.64 in.)

CYLINDER HEAD

DESCRIPTION	SPECIFICATION
Combustion Chamber	55.22 - 58.22 cc (3.37 - 3.55 cu. in.)
Valve Seat Angle	44.5°
Valve Seat Runout (MAX)	0.064 mm (0.0025 in.)
Valve Seat Width	1.02 - 1.52 mm (0.040 - 0.060 in.)
Valve Guide I. D (Integral)	7.95 - 7.97 mm (0.313 - 0.314 in.)
Valve Stem to Guide Clearance	0.025 - 0.076 mm (0.001 - 0.003 in.)
Cylinder Head Warpage (Flatness)	0.03 mm per 25 mm (0.001 in. per 1 in.) 0.05 mm per 152 mm (0.002 in. per 6 in.)
Flatness (MAX)	0.20 mm - max for total length

OIL PUMP

DESCRIPTION	SPECIFICATION
Gear to Body Clearance (Radial)	0.051 - 0.102 mm (0.002 - 0.004 in.)
Gear to Body Clearance (Radial Preferred)	0.051 mm (0.002 in.)
Gear End Clearance (Plastigage)	0.051 - 0.152 mm (0.002 - 0.006 in.)
Gear End Clearance (Plastigage Preferred)	.051 mm (.002 in.)
Gear End Clearance (FeelerGauge)	0.1016 - 0.2032 mm (0.004 - 0.008 in.)
Gear End Clearance (FeelerGauge Preferred)	0.1778 mm (0.007 in.)